DIFFERENTIALS OF A PRODUCT OF ALGEBRAIC SPACES SPLIT AS A DIRECT SUM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1226-1240 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1242-1247, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The statement identifies the domain as the direct sum indexed by the two projections.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The functoriality maps c_f (lemma-functoriality-differentials), etale-locality of the claim (lemma-localize-differentials), and the schemes case (Morphisms, lemma-differential-product).

Source reference. Original lemma 04CX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1062–1083. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1247–1247. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ and $g : Y \to B$ be morphisms of algebraic spaces over S with the same target. Let $p : X \times_B Y \to X$ and $q : X \times_B Y \to Y$ be the projection morphisms. The maps from Lemma 04CX*

$$p^*\Omega_{X/B} \oplus q^*\Omega_{Y/B} \longrightarrow \Omega_{X \times_B Y/B}$$

give an isomorphism.

Proof. Follows from the schemes version, see Morphisms, Lemma 01V1 and étale localization, see Lemma 04CU. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.